Highlights

Pitfalls and Benchmarking of Metaheuristic-Tuned Nonlinear Controllers in Robotic Systems: Equal-Budget Validation, Digital Limits, and Generalization

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- Equal-budget, paired statistical benchmark (NFE=2000, 30 runs) of GWO, mGWO, PSO, and $ACO_{\mathbb{R}}$.
- Nominal plant tuning drives gains to the digital stability limit (49.6 Hz chatter at 100 Hz sampling).
- Gains tuned solely on the nominal model fail by 4–5 orders of magnitude under parameter shifts.
- Mechanistic ablation reveals hybrid escape mechanisms never activate or add no statistical advantage.
- Demonstrates that hyperparameter-budget alignment outweighs algorithmic over-engineering in $D = 4$.

Pitfalls and Benchmarking of Metaheuristic-Tuned Nonlinear Controllers in Robotic Systems: Equal-Budget Validation, Digital Limits, and Generalization

J. E. Cervantes-Reyes[a,i], Ma. de los Angeles Alamilla Daniel[a] and Angel Ricardo Licona Rodríguez[a]

[a] *Universidad Politécnica de Pachuca, Carretera Pachuca-Cd. Sahagún km 20, Ex-Hacienda de Santa Bárbara, Zempoala, 43830, Hidalgo, Mexico*

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ABSTRACT

Metaheuristics are widely used to tune nonlinear controllers for robotic manipulators and rehabilitation devices. However, prevailing benchmarking practices suffer from two critical flaws: evaluating the optimizer exclusively on the nominal plant model used for control design, and comparing algorithms under unequal computational budgets. Furthermore, tuned gains are rarely evaluated across unseen perturbed plants. We propose a rigorous validation protocol for REHAP, a two-degree-of-freedom lower-limb rehabilitation robot driven by backstepping control. The protocol trains each independent run on a distinct perturbed plant and evaluates performance across 20 unseen plants subjected to neuromuscular disturbance pulses. Under a strictly equal budget of 2000 cost evaluations (NFE) across 30 paired runs, a modified Grey Wolf Optimizer equipped with four escape mechanisms (mGWO), canonical GWO, PSO, and $ACO_{\mathbb{R}}$ achieve statistically indistinguishable validation tracking quality (Friedman $p = 0.65$, all Holm-adjusted pairwise $p \geq 0.79$). Notably, PSO requires significantly more evaluations to attain its convergence plateau ($p = 0.0014$). Mechanistic activation counters reveal that one escape mechanism in mGWO never activates and its exploration parameter fails to reach zero within budget. A schedule-corrected variant (mGWO-a) restores benchmark precision on synthetic landscapes (Ackley median $0.108 \rightarrow 0.0094$) but provides no benefit for controller tuning, with mGWO never outperforming canonical GWO. Crucially, nominal tuning drives gains toward the digital stability boundary (49.6 Hz oscillations at 100 Hz sampling) and induces severe actuator saturation under perturbations. Our findings demonstrate that in low-dimensional controller tuning, proper budget-parameter alignment and digital limit enforcement matter far more than the ad-hoc addition of metaheuristic escape mechanisms.

1. Introduction

Lower-limb rehabilitation robots must execute pre-planned therapeutic trajectories with high precision despite unpredictable patient-robot interactions and parameter uncertainties. Model-based nonlinear controllers such as backstepping [15, 17] exploit the Euler–Lagrange structure of manipulators [27], and recent implementations often pair backstepping with disturbance estimation to mitigate parameter uncertainty [21, 22]. However, closed-loop tracking precision and transient stability heavily depend on selecting control gains. Because manual trial-and-error tuning is tedious, non-reproducible, and frequently sub-optimal, derivative-free metaheuristics have become the tool of choice [1, 8]. Representative algorithms include the Grey Wolf Optimizer (GWO) [19], Particle Swarm Optimization (PSO) [6, 14, 24], and Continuous Ant Colony Optimization ($ACO_{\mathbb{R}}$) [10, 25]. A proliferation of hybrid GWO variants have introduced multi-strategy initializations, Cauchy mutations, adaptive restarts, and local search routines to overcome premature convergence [2, 7, 20].

Despite hundreds of publications on metaheuristic-tuned controllers, two pervasive methodological flaws undermine the validity of comparative claims:

1. *The nominal evaluation fallacy:* The optimizer searches for gains using the exact, unperturbed nominal plant model, and the tuned controller is then validated on that same nominal model [3, 26]. The critical engineering question—whether the resulting gains remain stable and effective on plants that deviate from the design model—is routinely ignored.
2. *Unequal computational budgets:* Algorithms are frequently compared across arbitrary iteration counts without standardizing the number of function evaluations (NFE), giving hybridized algorithms an unfair computational advantage [16, 18].

[i] Corresponding author
cervantesreyesjesusedg16@gmail.com (J. E. Cervantes-Reyes); alamillameca@upp.edu.mx (Ma. de los Angeles Alamilla Daniel); arliconar@upp.edu.mx (Angel Ricardo Licona Rodríguez)
ORCID(s): 0009-0004-9457-3162 (J. E. Cervantes-Reyes)

In robot-aided physical therapy, patient morphology (limb mass, segment inertia, muscle tone) varies substantially between individuals. Moreover, involuntary neuromuscular contractions such as spasticity, clonus, or tremors inject abrupt torque and position disturbances during rehabilitation exercises. Controllers tuned purely on nominal dynamics fail catastrophically under these physical conditions.

Contributions This paper establishes a rigorous benchmarking methodology for metaheuristic-tuned nonlinear controllers, utilizing the REHAP two-degree-of-freedom lower-limb rehabilitation robot as a case study:

1. *A robust validation protocol:* An optimization framework that trains on perturbed plants and independently validates tuned gains across 20 unseen plants under distinct neuromuscular pulse disturbance patterns (Sec. 4).
2. *Discovery and characterization of digital stability limits:* Analytical and numerical evidence showing that nominal tuning pushes control gains to the Nyquist limit of sampled-data zero-order-hold (ZOH) control (49.6 Hz oscillations at 100 Hz sampling rate), inducing severe voltage saturation under real-world perturbations (Sec. 6A–B).
3. *A strictly equal-budget, paired statistical benchmark:* A rigorous comparison of a modified GWO (mGWO), schedule-corrected mGWO-a, canonical GWO, PSO, and $ACO_{\mathbb{R}}$ across 30 paired runs with Holm-adjusted Wilcoxon and Vargha–Delaney A_{12} statistics (Sec. 6.4).
4. *Mechanistic deconstruction of hybrid metaheuristic operators:* Empirical tracking of activation counters and ablation showing that complex escape mechanisms fail to activate or add no measurable benefit, formulating hypothesis H^{i} on the primacy of hyperparameter-budget alignment over algorithmic over-engineering in low-dimensional controller tuning (Sec. 6.5–6.5).

2. Related Work and State-of-the-Art Gap

Backstepping provides a systematic Lyapunov design for strict-feedback systems [15, 17]. For fully actuated robotic manipulators, backstepping yields an inverse-dynamics control law with a filtered tracking error surface (Sec. 3.2). In controller tuning, metaheuristics balance exploration and exploitation [7]. GWO achieves this balance through a linearly decaying control parameter a, PSO via inertia weights and stochastic acceleration coefficients, and $ACO_{\mathbb{R}}$ through a Gaussian mixture archive [6, 19, 25]. Hybrid and memetic extensions combine these population dynamics with local search operators [2, 20].

The metaheuristics community has extensively critiqued metaphor-based algorithms [3, 26] and benchmarks centered at the coordinate origin, which introduce structural search biases [16, 18]. Furthermore, implementation choices such as bound-handling mechanisms often dominate algorithmic behavior [12]. Best practices dictate non-parametric hypothesis testing with family-wise error rate control (Holm correction [13]) and non-parametric effect sizes (A_{12} [28]) across identical evaluation budgets [9]. In accordance with the No-Free-Lunch theorem [29], no single optimizer should be expected to universally dominate across diverse objective landscapes.

Recent studies on metaheuristic-tuned nonlinear controllers for lower-limb exoskeletons [11, 30] and manipulators [1, 8, 23] consistently tune and validate controllers on the nominal plant or on a single static uncertainty scenario, omitting equal-budget multi-algorithm comparisons. While scenario-based robust tuning has been explored for inverted pendulums [31], it was restricted to a single PID family without multi-optimizer statistical benchmarking. Table 1 summarizes this critical gap in the recent literature.

3. System Dynamics and Backstepping Control

3.1. Euler–Lagrange robot dynamics

REHAP is a mechatronic test bench developed for lower-limb physical therapy (hip and knee flexion/extension; see [4, 5]) with two revolute joints $q = [q_1, q_2]$ driven by direct-drive DC servomotors ($r_i = 1$). The rigid-body Euler–Lagrange dynamics with reflected motor inertia are

$$.M.q/ + J_m/q + C.q, q/q + G.q/ + Fq = \tau, \quad (1)$$

where $F = \text{diag}.f_{v1}, f_{v2}/$, $J_m = \text{diag}.J_{m1}, J_{m2}/$, and the elements of the symmetric inertia matrix $M.q/$ are $M_{11} = m_1 l_{c1}^2 + m_2.l_1^2 + l_{c2}^2 + 2l_1 l_{c2} \cos q_2/ + I_1 + I_2$, $M_{12} = M_{21} = m_2.l_{c2}^2 + l_1 l_{c2} \cos q_2/ + I_2$, and $M_{22} = m_2 l_{c2}^2 + I_2$. With $h = m_2 l_1 l_{c2} \sin q_2$, the Coriolis matrix is

$$C.q, q/ = \begin{bmatrix} *hq_2 & *h.q_1 + q_2/ \\ hq_1 & 0 \end{bmatrix}, \quad (2)$$

and the gravitational torque vector is

$$G.q/ = \begin{bmatrix} .m_1 l_{c1} + m_2 l_1/g \cos q_1 + m_2 g l_{c2} \cos.q_1 + q_2/ \\ m_2 g l_{c2} \cos.q_1 + q_2/ \end{bmatrix}. \quad (3)$$

For the nominal parameters listed in Table 2, $\det M.q/ \geq 2.4 \,\dot{\text{I}}\, 10^{*3}$ over $q_2 \,\ddot{\text{E}}\, [*\pi, \pi]$, ensuring that $M.q/$ remains uniformly positive definite. Moreover, $M * 2C$ is skew-symmetric, satisfying standard passivity properties [27].

Table 1
Metaheuristic-tuned nonlinear controllers for manipulators/exoskeletons: validation and comparison practices in recent literature.

Study	Application	Optimizer(s)	Train/validate on different plants	Equal-budget, multi-optimizer stats
[1]	3-link flexible-joint manipulator	GA-tuned SMC	No (nominal only)	No (single optimizer)
[8]	Underwater vehicle-manipulator	GWO	No (nominal only)	No (single optimizer)
[11]	Lower-limb rehab. exoskeleton	GWO	No (nominal only)	No (single optimizer)
[30]	Lower-limb rehab. exoskeleton	GWO + fuzzy SMC	No (nominal only)	No (single optimizer)
[23]	Robotic manipulator	Extended GWO	No (nominal only)	No (single optimizer)
[31]	Cart-inverted pendulum	(FO)PID, scenario-tuned	Yes (uncertainty scenarios)	No (single optimizer family)
This work	REHAP lower-limb rehab. robot	mGWO, GWO, PSO, $ACO_{\mathbb{R}}$	Yes (20 unseen perturbed plants)	Yes (paired, equal NFE, Friedman/Holm/A_{12})

Table 2
Nominal parameters of the REHAP rehabilitation robot (SI units).

Parameter	Value	Parameter	Value	Parameter	Value
m_1, m_2	7.0, 1.0 kg	l_1	0.50 m	l_{c1}, l_{c2}	0.186, 0.10 m
I_1, I_2	10^{*5}, $4\dot{1}0^{*5}$ kg m^2	f_{v1}, f_{v2}	0.05 N m s/rad	J_{m1}, J_{m2}	0.062 kg m^2
$K_a = K_b$	1.6206 N m/A	R_a	1.8 Ω	$v_{\max}, \tau_{\max}$	24 V, 40 N m

Neglecting armature inductance (as electrical time constants are substantially faster than rigid-body dynamics, $\tau_e \ ^{\rightarrow} \ \tau_m$), the DC motor dynamics reduce to the algebraic torque-voltage relation:

$$\tau_i = \frac{K_a}{R_a} v_i * \frac{K_a K_b}{R_a} q_i, \qquad v_i \le v_{\max} = 24 \text{ V}, \quad (4)$$

which represents the physical actuation limit of the test bench ($\tau_{\max} = 40$ N m is the nominal structural limit).

The therapeutic reference trajectory is a smooth quintic polynomial over $T = 10$ s moving from $q_1 = {}^{*}\pi\dot{}2$ to $\pi\dot{}4$ and $q_2 = 0$ to $\pi\dot{}4$ rad, with zero initial and terminal velocities and accelerations. Numerical integration is performed using Runge–Kutta 4th order (RK4) with zero-order-hold digital control at $T_s = 0.01$ s (100 Hz).

3.2. Backstepping controller formulation

Defining position tracking error $e = q \ * \ q_d$, gain matrices $K_1 = \text{diag}.k_{11}, k_{12}/$ and $K_2 = \text{diag}.k_{21}, k_{22}/$, virtual stabilizing control $\alpha = q_d \ * \ K_1 e$, and velocity error surface $z = q \ * \ \alpha = e + K_1 e$, the commanded backstepping torque is synthesized as

$$\tau^c = M.q/.q_d {}^{*} K_1 e {}^{*} K_2 z/ + C.q, q/q + G.q/ + Fq. \quad (5)$$

Motor armature voltages v_i are computed by inverting (4) while injecting $J_m q_r$ ($q_r = q_d \ * \ K_1 e$) to cancel reflected rotor inertia. Under nominal dynamics without saturation, the closed-loop error system satisfies

$$z = {}^{*}\Lambda.q/K_2 z, \qquad e = {}^{*}K_1 e + z, \qquad \Lambda = .M + J_m/^{*1} M. \quad (6)$$

Because $\Lambda.q/$ is similar to a symmetric positive definite matrix, its eigenvalues are real and strictly contained in $.0, 1/$. Along the prescribed trajectory, numerical evaluation yields $\lambda.\Lambda.q//$ Ë $[0.061, 0.908]$. The slow direction decays at roughly $6\tilde{}$ of the rate nominal K_2 suggests, explaining why optimizers tend to drive k_{22} to large values.

Proposition 1. ***(Nominal exponential stability, sufficient condition).*** *For the nominal, continuous-time, unsaturated error dynamics (6), define, for each q along the reference trajectory,*

$$\lambda_2.q/ := \lambda_{\min}\big(\tfrac{1}{2}.\Lambda.q/K_2 + K_2\Lambda.q///, \quad (7)$$

as the minimum eigenvalue of the symmetric part of $\Lambda.q/K_2$. If $\lambda_{\min}.K_1/ > \frac{1}{2}$ and $\inf_q \lambda_2.q/ > \frac{1}{2}$ along the trajectory, the origin $.e, z/ = 0$ is exponentially stable.

Proof. Consider the Lyapunov function candidate $V = \frac{1}{2}e^2 + \frac{1}{2}z^2$. Taking the time derivative along trajectories of (6) yields $V = ee + zz = {}^{*}eK_1 e + ez \ {}^{*} \ z\Lambda K_2 z$. Because quadratic forms depend solely on the symmetric part of their kernel matrix, $z\Lambda K_2 z \ge \lambda_2.q/z^2$ by the Rayleigh–Ritz theorem. Applying Cauchy–Schwarz and Young's inequality ($ez \le \frac{1}{2}e^2 + \frac{1}{2}z^2$),

$$V \le {}^{*}\big(\lambda_{\min}.K_1/ \ * \ \tfrac{1}{2}/e^2 \ * \ \big(\lambda_2.q/ \ * \ \tfrac{1}{2}/z^2 \le {}^{*}2cV,$$

where $c = \min\hat{}\lambda_{\min}.K_1/ \ * \ \frac{1}{2}, \ \inf_q \lambda_2.q/ \ * \ \frac{1}{2}\text{‘} > 0$. Consequently, $V.t/ \le V.0/e^{*2ct}$.

Remark 1. Over the full operational workspace $q_2 \ddot{E} [*\pi, \pi]$, numerical evaluation confirms that for typical optimized gains (e.g., GWO run 19, $K_2 = \text{diag}.26.1, 78.4/$), $\lambda_2.q/ \geq 4.41 > 0.5$ and $\lambda_{\min}.K_1/ > 0.5$ hold strictly across the workspace, guaranteeing continuous exponential stability. Conversely, near the lower search bound $lb = [1, 1, 1, 1]$, $\inf_q \lambda_2.q/ \grave{u}\ 0.061$, showing that while the sufficient condition fails for arbitrarily small gains, the optimizers reliably identify gains well inside the stable basin. Note that Proposition 1 assumes continuous-time, unsaturated operation; physical sampled-data and voltage saturation limits are governed by the digital limit analyzed in Sec. 6.2.

The parameter vector to tune is $\theta = [k_{11}, k_{12}, k_{21}, k_{22}] \ddot{E} \mathbb{R}^4$.

4. Cost Function, Disturbance Model, and Validation Protocol

4.1. Tracking objective and penalization

For any candidate gain vector θ, closed-loop dynamics are simulated for $T = 10$ s. The scalar objective function is

$$J.\theta/ = \text{RMS}_1 + \text{RMS}_2 + \lambda_{\tau i} \overline{\max.0, \tau_i^c * \tau_{\max}/^2} + \lambda_{sat} f_{sat}, \tag{8}$$

where $\text{RMS}_i = \sqrt{\frac{1}{K} \sum_{k=1}^{K} .q_i.t_k/ * q_{di}.t_k//^2}$ denotes the true position root-mean-square tracking error, τ^c is the unconstrained commanded torque from (5), the overbar denotes temporal averaging, f_{sat} represents the fraction of time steps where motor voltage reaches saturation ($v_i > 0.99 v_{\max}$), and weights are fixed at $\lambda_\tau = 1$ and $\lambda_{sat} = 10$. Any simulation exhibiting numerical divergence or instability is assigned a penalty of $J = 10^6$.

4.2. Morphological uncertainty and neuromuscular pulses

Parametric uncertainty is injected into all physical parameters $\vartheta_k \ddot{E} \hat{} m_1, m_2, l_{c1}, l_{c2}, I_1, I_2, f_{v1}, f_{v2}$‘:

$$\vartheta_k = \vartheta_k.1 + \rho u_k/, \qquad u_k \acute{\imath} \boldsymbol{\mathcal{U}}.*1, 1/, \quad \rho = 0.15, \tag{9}$$

representing inter-subject morphological variations (,15˜) encountered in clinical practice, while the controller retains the nominal parameters ϑ_k.

In rehabilitation robotics, patients frequently exhibit spastic hypertonia, muscle spasms, or tremor. To model these abrupt joint torque/position disturbances, an external pulse train is added to measured feedback:

$$q^m.t/ = q.t/ + p.t/,$$
$$p.t/ = \begin{cases} A & t \geq t_0,\ .t * t_0/ \bmod P < w \\ 0 & \text{otherwise,} \end{cases} \tag{10}$$

with amplitude $A = 0.01$ rad, period $P = 2$ s, pulse width $w = 0.05$ s, and onset $t_0 = 1$ s. In validation, an alternative, more aggressive spastic disturbance is also evaluated: $A = 0.02$ rad, $P = 2.5$ s, $w = 0.1$ s, and $t_0 = 0.5$ s.

4.3. Optimization problem and benchmarking protocol

Three tuning environments are defined: (A) nominal plant without pulses; (B) a single perturbed plant with disturbance pulses; (C) an ensemble mean cost over five fixed perturbed plants:

$$J.\theta/ = \tfrac{1}{5} \sum_{s=1001}^{1005} J.\theta; \text{plant}_s/. \tag{11}$$

The bounded optimization problem is

$$\theta^\star = \arg \min_{\theta \ddot{E}[lb,ub]} J.\theta/, \qquad lb = [1, 1, 1, 1], \tag{12}$$

with $ub = [1000]^4$ in exploratory stages and $ub = [1000, 1000, 1000, 600]$ in the final protocol (justified in Sec. 6.2).

Because $J.\theta/$ contains non-differentiable kinks (voltage saturation, piecewise pulse disturbances) and severe step-penalty discontinuities (10^6) at the numerical stability boundary, gradient-based techniques fail to converge reliably. Derivative-free metaheuristics are budgeted strictly in terms of cost function evaluations (NFE). In the final confirmatory protocol, each independent run $k \ddot{E} \hat{} 1, \breve{g}, 30$‘ trains all optimizers on the exact same perturbed training plant (seed $= 3000 + k$, paired design). Validation is performed by testing the resulting gains across 20 unseen perturbed plants (seeds 2001–2020) under both training and alternative disturbance patterns.

5. Optimization Algorithms and Hyperparameters

All optimizers operate under a standardized budget of $NFE_{\max} = 2000$. Table 3 lists all hyperparameter values, kept invariant across all experiments.

Canonical GWO [19] A population of N wolves is guided by the three best individuals $(X_\alpha, X_\beta, X_\delta)$: $A_\updownarrow = 2a r_1 * a$, $C_\updownarrow = 2 r_2$, $X_\updownarrow^{\acute{I}} = X_\updownarrow * A_\updownarrow C_\updownarrow X_\updownarrow * X_i$, with $r_1, r_2 \acute{\imath} \boldsymbol{\mathcal{U}}.0, 1/^D$, and updated position $X_i \leftarrow \text{clip}..X_\alpha^{\acute{I}} + X_\beta^{\acute{I}} + X_\delta^{\acute{I}}/˙3/$. Parameter a decays linearly: $a = 2.1 * t˙T_{\max}/$, where $T_{\max} = .NFE_{\max} * N/˙N$. Elitism is enforced by archiving the overall best solution found.

Modified GWO (mGWO) and Corrected mGWO-a Built upon GWO, mGWO integrates four hybrid mechanisms:

1. *Informed seeding:* $n_s = 5$ individuals are initialized from manual baseline gains $[5, 5, 4, 4]$ and

Algorithm 1 mGWO (italicized text denotes added hybrid mechanisms)

Require: $N, NFE_{\max}, [lb, ub], n_s, \epsilon, c_{mut}, c_{res}, \phi, n_{mut}, n_{loc}$

Initialize n_s agents from informed seeds; $N * n_s$ uniformly
Evaluate all agents; $NFE \leftarrow N$; $t \leftarrow 0$; $c \leftarrow 0$
while $NFE < NFE_{\max}$ **do**
  Rank agents; set $X_\alpha, X_\beta, X_\delta$; $a \leftarrow 2.1 * t \cdot T_{\max}/$
  for each agent $i = 1, \breve{g}, N$ **do**
    $X_i \leftarrow \text{clip}\big(.X_\alpha^{\hat{I}} + X_\beta^{\hat{I}} + X_\delta^{\hat{I}}/\cdot 3/$
  end for
  Evaluate updated agents; $NFE \leftarrow NFE + N$
  $c \leftarrow 0$ if $J.X_\alpha/$ improved by $\geq \epsilon$, else $c \leftarrow c + 1$
  if $c \geq c_{res}$ **then**
    Recovery: reinitialize worst ϕN agents; $c \leftarrow 0$
  else if $c \geq c_{mut}$ **then**
    Mutation: perturb worst n_{mut} agents
  end if
  Local search: sample n_{loc} points around X_α at radius r_{loc}; replace X_α if improved
  $NFE \leftarrow NFE+$ (extra evaluations); $t \leftarrow t + 1$
end while
return Best archived solution

structured perturbations; the remaining $N * n_s$ are initialized uniformly.

2. *Stagnation mutation:* A stagnation counter c increments if leader cost does not improve by at least $\epsilon = 10^{*6}$. When $c \geq c_{mut} = 2$, the $n_{mut} = 10$ worst individuals undergo Gaussian mutation ($\sigma = 0.05.ub * lb/$).
3. *Stagnation recovery:* If stagnation reaches $c \geq c_{res} = 5$, a fraction $\phi = 0.2$ of the worst population is re-initialized uniformly over $[lb, ub]$.
4. *Elitist local search:* In every iteration, $n_{loc} = 5$ candidates are sampled within a hypersphere of radius $r_{loc} = r_0.1 * t \cdot T_{\max}/^{\gamma}$ around X_α, replacing X_α only if superior.

Crucially, mechanisms 2–4 consume additional function evaluations per iteration. In mGWO, $T_{\max}$ is calculated assuming only population updates (N evals/iter), causing the budget to exhaust before t reaches $T_{\max}$, leaving parameter a and local search radius r_{loc} truncated at high exploration values. The corrected variant, mGWO-a, computes progress using actual evaluation consumption: $NFE \cdot NFE_{\max}$. Algorithm 1 summarizes the execution flow.

PSO [6, 14] Standard velocity update $v \leftarrow wv + c_1 r_1 .p_i * x/ + c_2 r_2 .g * x/$, with inertia weight w decaying linearly from 0.9 to 0.4, velocity clamping $v \leq 0.2.ub*lb/$, and reflection at bounds with velocity inversion.

$ACO_{\mathbb{R}}$ [25] Continuous Ant Colony Optimization maintaining an archive of $k = 50$ solutions. In each iteration, $m = 5$ ants construct solutions via Gaussian mixture sampling centered on ranked archive solutions with selection weights w_l and coordinate deviations proportional to average inter-solution distances.

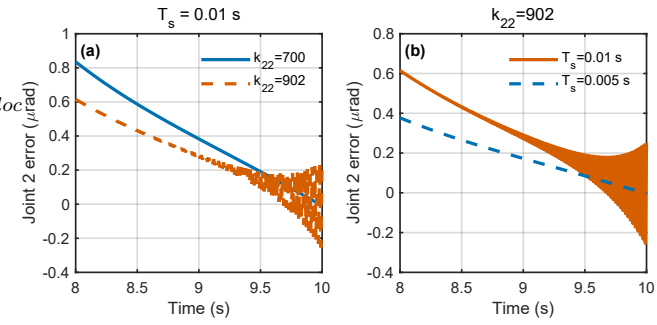


Figure 1: Joint-2 tracking error in $[8, 10]$ s under nominal plant simulation (μrad). (a) $T_s = 0.01$ s: smooth tracking for $k_{22} \leq 700$, high-frequency chatter at 49.6 Hz for k_{22} ù 902. (b) Halving T_s to 0.005 s suppresses the Nyquist oscillation, confirming a digital ZOH stability boundary.

6. Experimental Results and Discussion

6.1. The nominal tuning fallacy

Optimizing on the unperturbed nominal plant across 100–200 iterations leads all four algorithms to converge to virtually identical cost values (J ù 4.24 İ 10^{*6} to 4.34 İ 10^{*6}, differences $< 1\tilde{}$). Because the backstepping controller encapsulates exact model parameters, arbitrarily large gains yield near-zero simulation errors. Consequently, the nominal plant is fundamentally incapable of discriminating optimizer efficacy.

6.2. Identification of the digital stability limit

Nominal tuning consistently pushes k_{22} toward the upper bound (k_{22} ù 900). Fig. 1 analyzes joint-2 tracking error as a function of k_{22} and sampling step T_s. For $k_{22} \leq 700$, error remains perfectly smooth (peak-to-peak < 3.6 İ 10^{*8} rad). However, at k_{22} ù 902, high-frequency chatter erupts with a dominant spectral peak at exactly 49.6 Hz—matching the Nyquist frequency of the 100 Hz digital loop ($f_{Nyquist} = 1\cdot.2T_s/ = 50$ Hz). Halving the sampling period to $T_s = 0.005$ s entirely eliminates this chatter, confirming that the instability is a zero-order-hold discretization artifact. At $k_{22} = 1000$, the discrete loop becomes unstable. Thus, nominal optimization drives gains directly onto the digital stability boundary. To prevent discretization-induced instability, we enforce $ub.k_{22}/ = 600$ in all subsequent protocols.

6.3. Fragility of nominal gains under plant perturbations

When gain sets tuned exclusively on the nominal plant are validated across 20 unseen perturbed plants

Table 3
Hyperparameter configurations across all optimizers (strictly standardized).

Algorithm	Parameters
GWO	$N = 20$
mGWO, mGWO-a	$N = 20, n_s = 5, \epsilon = 10^{*6}, c_{mut} = 2, c_{res} = 5, \phi = 0.2, n_{mut} = 10, \sigma = 0.05, n_{loc} = 5, r_0 = 0.1, \gamma = 2$
PSO	$N = 20, c_1 = c_2 = 2.0, w_{\max} = 0.9, w_{\min} = 0.4, v_{\max} = 0.2.ub * lb/$, reflection bounds
$ACO_{\mathbb{R}}$	$k = 50, m = 5, q = 0.1, \xi = 0.85$

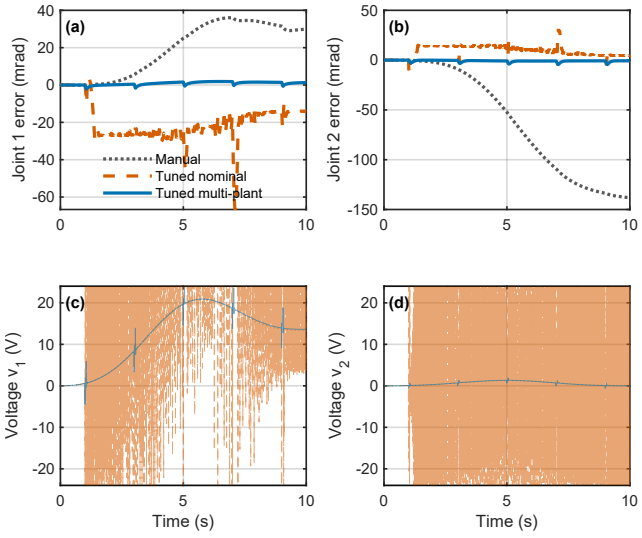


Figure 2: Closed-loop response on an unseen perturbed plant for manual, nominal-tuned, and multi-plant-tuned gains. (a,b) Tracking errors (mrad); (c,d) Armature voltages (,24 V limits). Nominal gains chatter between saturation boundaries; multi-plant gains maintain smooth tracking within ù 10 mrad without saturation.

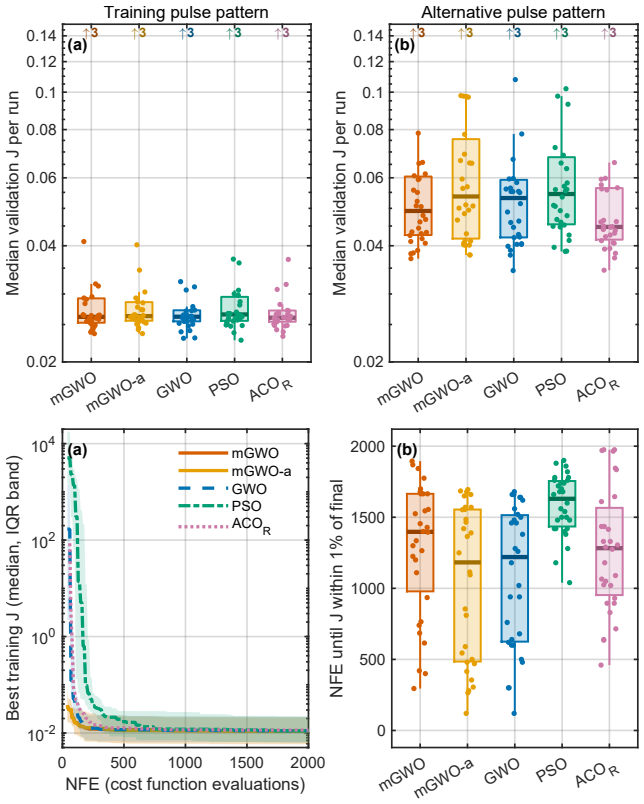


Figure 3: Confirmatory benchmarking over 30 paired runs (log scale). (a) Median validation cost under training disturbance pulses; (b) Validation cost under alternative pulses; (c) Convergence of best training cost vs. NFE (interquartile bands); (d) Evaluations required to reach within 1% of final cost (NFE_1-). All optimizers achieve indistinguishable validation quality; PSO converges significantly slower.

with disturbance pulses, performance collapses catastrophically: validation cost median spikes to $1.3 \dot{I}\, 10^2$–$6.4 \dot{I} 10^3$, worst-case cost exceeds 10^6, and motor voltages saturate during 7–79˜ of the trajectory. In stark contrast, gains tuned under perturbed plants achieve validation costs of ù 0.026 (four to five orders of magnitude superior). Fig. 2 illustrates this mechanism: overly aggressive nominal gains amplify disturbance pulses, triggering severe voltage saturation and high-frequency limit cycles.

6.4. Confirmatory protocol: 30 paired runs

Table 4(a) and Fig. 3 present the results of the 30 paired runs under the final validation protocol ($NFE_{\max} = 2000$).

Solution quality equivalence Across all 30 paired runs, validation costs between the five optimizers are statistically indistinguishable (medians 0.02603–0.02655, spread < 2˜). The non-parametric Friedman test reveals no omnibus difference ($p = 0.65$), and all pairwise Wilcoxon tests with Holm adjustment yield $p \geq 0.79$, with Vargha–Delaney effect sizes concentrated at parity (A_{12} Ë [0.45, 0.55]). Evaluating on the alternative pulse pattern confirms invariance ($p = 0.24$). Schedule-corrected mGWO-a does not differ from baseline mGWO ($p = 1.0, A_{12} = 0.46$).

Heavy-tail vulnerability Every optimizer exhibits a heavy right tail in validation: 90th percentiles cluster at 0.70–0.72, and worst-case costs reach 509–628. This proves that optimizing average cost J fails to protect against extreme saturation events, which stem from plant uncertainty rather than optimizer deficiencies.

Table 4
(a) Final validation protocol (30 paired runs, $NFE_{\max} = 2000$): median validation cost on 20 unseen plants under training (J_{val}) and alternative (J_{val}^{alt}) disturbance pulses, interquartile range (IQR), worst-case cost, and evaluations to $5^{\tilde{}}$ and $1^{\tilde{}}$ convergence plateaus. (b) Generalization check on a cubic Bézier trajectory ($T = 5$ s) without re-tuning.

(a) Final Validation Protocol

Algorithm	J_{val}	IQR	J_{val}^{alt}	Worst Case	$NFE_{5^{\tilde{}}}$	$NFE_{1^{\tilde{}}}$
mGWO	0.02616	0.0040	0.0492	509	510	1398
mGWO-a	0.02628	0.0030	0.0537	582	403	1183
GWO	0.02619	0.0017	0.0532	628	350	1220
PSO	0.02655	0.0039	0.0545	550	1320	1630
$ACO_{\mathbb{R}}$	0.02603	0.0018	0.0448	577	645	1283

(b) Generalization Check on Bézier Trajectory

Gain Configuration	RMSE q_1 (rad)	RMSE q_2 (rad)	ITSE ($\dot{I}10^{*5}$)	Voltage Saturation (%)
Manual $[5, 5, 4, 4]$	$2.35\dot{I}10^{*3}$	$7.38\dot{I}10^{*3}$	4.98	0.00
mGWO (run 2)	$1.14\dot{I}10^{*4}$	$9.36\dot{I}10^{*5}$	0.011	0.00
GWO (run 19)	$1.30\dot{I}10^{*4}$	$9.01\dot{I}10^{*5}$	0.015	0.00

Table 5
Additional scenarios: median final cost (Holm-adjusted p-values vs. mGWO in parentheses; bold indicates $p < 0.05$).

Algorithm	S1 (REHAP, 300 ev.)	S2 (REHAP, 600 ev.)	S3 (Shifted Rastrigin)	S4 (Shifted Ackley)
mGWO	0.0407	0.0402	2.98	0.108
mGWO-a	0.0414 (0.35)	0.0403 (0.47)	**2.00 (0.0095)**	**0.0094** ($3\dot{I}10^{*11}$)
GWO	0.0407 (0.68)	0.0394 (0.91)	2.99 (0.60)	**0.0083** ($8.5\dot{I}10^{*9}$)
PSO	**0.0490 (0.017)**	**0.0420 (0.031)**	**1.17** ($1.9\dot{I}10^{*5}$)	$4.8\dot{I}10^{*4}$ ($3\dot{I}10^{*11}$)
$ACO_{\mathbb{R}}$	0.0443 (0.43)	0.0370 (0.47)	**5.21** ($1.3\dot{I}10^{*5}$)	$2.1\dot{I}10^{*8}$ ($3\dot{I}10^{*11}$)

Convergence rate disparity While asymptotic quality is identical, convergence speed differs significantly (Friedman $p = 0.0014$ on $NFE_{1^{\tilde{}}}$). As shown in Fig. 3(c)-(d), PSO requires substantially more evaluations (1630) than GWO (1220), mGWO (1398), and $ACO_{\mathbb{R}}$ (1283) to reach its convergence plateau (Holm-adjusted p-values 0.001–0.042).

6.5. Benchmark landscapes and mechanistic deconstruction

Table 5 details performance across synthetic landscapes (shifted Rastrigin and Ackley in $D = 4$, with optima offset from the origin to eliminate center bias) and tight evaluation budgets (300 and 600 NFE).

On shifted Ackley, canonical GWO significantly outperforms mGWO (0.0083 vs. 0.108, $p = 8.5\,\dot{I}\,10^{*9}$). Activation counters explain this counterintuitive result: in mGWO, local search consumed $18.5^{\tilde{}}$ of all evaluations without finding improvements, and the stagnation recovery mechanism never triggered (0 activations across all runs because $c < c_{res}$). Furthermore, because $T_{\max}$ ignored the evaluation drain of local search, parameter a terminated at a ù 0.51 instead of 0, leaving residual exploration that crippled final convergence. Schedule-corrected mGWO-a fully restores benchmark precision ($0.108 \rightarrow 0.0094$), matching canonical GWO.

These findings support working hypothesis H^{i}: In low-dimensional controller tuning ($D = 4$), hyperparameter-budget alignment and boundary handling exert far greater influence on tracking performance than the inclusion of complex escape mechanisms.

7. Conclusions

This study presented a rigorous validation protocol and equal-budget statistical comparison for metaheuristic-tuned backstepping controllers on a lower-limb rehabilitation robot. The investigation establishes three key conclusions:

1. *Nominal tuning is hazardous:* Optimizing on the nominal plant drives gains directly into the digital Nyquist boundary (49.6 Hz chatter at 100 Hz sampling) and triggers catastrophic voltage saturation under model variations.
2. *Algorithmic equivalence:* Under strictly equal evaluation budgets (2000 NFE across 30 paired runs), mGWO, canonical GWO, PSO, and $ACO_{\mathbb{R}}$ achieve identical validation tracking quality (Friedman $p = 0.65$), with PSO converging significantly slower ($p = 0.0014$).
3. *The fallacy of hybrid over-engineering:* Adding hybrid escape mechanisms to GWO failed to improve controller tuning and degraded precision on

multimodal benchmarks due to internal parameter-budget mismatches.

Future work will explore risk-sensitive objective functions such as Conditional Value-at-Risk (CVaR) to suppress validation heavy tails, derive analytical digital stability bounds for sampled backstepping, and deploy the validated gain sets onto the physical REHAP test bench.

CRediT authorship contribution statement

J. E. Cervantes-Reyes: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft. **Ma. de los Angeles Alamilla Daniel:** Supervision, Methodology, Writing - review & editing. **Angel Ricardo Licona Rodríguez:** Conceptualization, Supervision, Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The complete MATLAB simulation codebase, optimization routines, paired test data, and analysis scripts supporting this study are publicly available in our reproducible GitHub repository at: https://github.com/UPP-Robotics/REHAP-Robust-Tuning.

CRediT authorship contribution statement

J. E. Cervantes-Reyes: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft. **Ma. de los Angeles Alamilla Daniel:** Supervision, Methodology, Writing - review & editing. **Angel Ricardo Licona Rodríguez:** Conceptualization, Supervision, Project administration, Writing - review & editing.

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